

APSC 2113 - Engineering Analysis I
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LQ5 - Fall 2026: Solutions and Explanations

Version A

Q1. $\int_{-\infty}^{\infty} e^x \delta(x) dx$

The delta fires at $x = 0$, which lies within $(-\infty, \infty)$. By the sifting property,

$$\int_{-\infty}^{\infty} e^x \delta(x) dx = e^0 = 1.$$

Answer: (A) 1

Q2. $\int_1^3 \cos(t) \delta(t - 2) dt$

The delta fires at $t = 2$, and $1 < 2 < 3$, so it is inside the limits. Evaluate the rest of the integrand there:

$$\int_1^3 \cos(t) \delta(t - 2) dt = \cos(2).$$

Note that the result is a number, not a function of t , which rules out choices (A) and (B).

Answer: (C) $\cos(2)$

Q3. $\int_{-\infty}^4 g(x) \delta(x - 5) dx$

The delta fires at $x = 5$, but the upper limit of integration is 4. Since $5 > 4$, the impulse lies outside the interval, so the integrand is zero everywhere on $(-\infty, 4]$:

$$\int_{-\infty}^4 g(x) \delta(x - 5) dx = 0.$$

Answer: (D) 0

Q4. *The integral of an impulse function is a step function.*

True. The running integral of $\delta(\tau)$ from $-\infty$ to t is 0 for $t < 0$ (the impulse has not yet occurred) and 1 for $t > 0$ (the impulse has been fully accumulated). This is exactly the unit step function $u(t)$.

Answer: (A) True

Q5. $\int_{-\infty}^{\infty} \delta(t) \cos(t) dt$

The delta fires at $t = 0$, inside the limits:

$$\int_{-\infty}^{\infty} \delta(t) \cos(t) dt = \cos(0) = 1.$$

Choice (C), $2 \cos(t)$, is not a number and does not follow from the sifting property.

Answer: (D) 1

Version B

Q1. $\int_{-\infty}^{\infty} \sin(x) \delta(x) dx$

The delta fires at $x = 0$, inside the limits:

$$\int_{-\infty}^{\infty} \sin(x) \delta(x) dx = \sin(0) = 0.$$

The integral is finite, so $\pm\infty$ are not possible; $\delta(x)$ has unit area, and the weight $\sin(0)$ is zero.

Answer: (D) 0

Q2. $\int_1^{10} \cos(t) \delta(t - 5) dt$

The delta fires at $t = 5$, and $1 < 5 < 10$, so it is inside the limits:

$$\int_1^{10} \cos(t) \delta(t - 5) dt = \cos(5).$$

Answer: (C) $\cos(5)$

Q3. $\int_{-\infty}^{-4} g(x) \delta(x) dx$

The delta fires at $x = 0$, but the upper limit is -4 . Since $0 > -4$, the impulse lies outside $(-\infty, -4]$, so the integral is

$$\int_{-\infty}^{-4} g(x) \delta(x) dx = 0.$$

Answer: (D) 0

Q4. *Step response is the integral of the impulse function.*

Per the answer key, this is **True**: integrating the impulse gives the unit step, $\int_{-\infty}^t \delta(\tau) d\tau = u(t)$. (Equivalently, for a linear time-invariant system, the step response is the running integral of the impulse response.)

Answer: (A) True

Note on wording: strictly speaking, “step response” refers to a system’s output when driven by a step input, while the *integral of the impulse function* is the unit step function itself. The statement is true under the intended reading (impulse $\rightarrow$ step), but you may want to reword it as “The unit step function is the integral of the impulse function” to avoid ambiguity.

Q5. $\int_{-\infty}^{\infty} \delta(t) \tan(t) dt$

The delta fires at $t = 0$, inside the limits:

$$\int_{-\infty}^{\infty} \delta(t) \tan(t) dt = \tan(0) = 0.$$

The sifting property only requires $\tan(t)$ to be continuous at $t = 0$, which it is (its singularities are at $t = \pm\pi/2, \dots$, far from the impulse).

Answer: (B) 0